

Circular Measure

ANSWER KEY



Converting between degrees and radians

- 1 Convert 135° to radians, in terms of π .

$$135^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{4}.$$

- 2 Convert $\frac{5\pi}{6}$ radians to degrees.

$$\frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ.$$

- 3 Convert 210° to radians, in terms of π .

$$210^\circ \times \frac{\pi}{180^\circ} = \frac{7\pi}{6}.$$

- 4 Convert $\frac{7\pi}{4}$ radians to degrees.

$$\frac{7\pi}{4} \times \frac{180^\circ}{\pi} = 315^\circ.$$

Arc length

- 5 A circle has radius 8 cm. Find the length of an arc subtending an angle of $\frac{\pi}{3}$ radians at the center, using $s = r\theta$.

$$s = r\theta = 8 \times \frac{\pi}{3} = \frac{8\pi}{3} \text{ cm}.$$

- 6 A circle has radius 12 cm. Find the length of an arc subtending an angle of $\frac{5\pi}{6}$ radians at the center, using $s = r\theta$.

$$s = r\theta = 12 \times \frac{5\pi}{6} = 10\pi \text{ cm}.$$

Sector area

- 7 Find the area of a sector with radius 6 cm and angle $\frac{2\pi}{3}$ radians, using $A = \frac{1}{2}r^2\theta$.

$$A = \frac{1}{2}(6)^2\left(\frac{2\pi}{3}\right) = \frac{1}{2}(36)\left(\frac{2\pi}{3}\right) = 12\pi \text{ cm}^2.$$

- 8 A sector has radius 10 cm and area 40 cm^2 . Find the angle θ in radians.

$$40 = \frac{1}{2}(10)^2\theta = 50\theta, \text{ so } \theta = \frac{40}{50} = 0.8 \text{ radians}.$$

- 9 Find the area of a sector with radius 9 cm and angle $\frac{\pi}{6}$ radians, using $A = \frac{1}{2}r^2\theta$.

$$A = \frac{1}{2}(9)^2\left(\frac{\pi}{6}\right) = \frac{1}{2}(81)\left(\frac{\pi}{6}\right) = \frac{81\pi}{12} = \frac{27\pi}{4} \text{ cm}^2.$$

Perimeter of a sector

- 10 Find the perimeter of a sector with radius 7 cm and angle 1.2 radians.

$$\text{Perimeter} = 2r + r\theta = 2(7) + 7(1.2) = 14 + 8.4 = 22.4 \text{ cm.}$$

- 11 Find the perimeter of a sector with radius 5 cm and angle 2 radians.

$$\text{Perimeter} = 2r + r\theta = 2(5) + 5(2) = 10 + 10 = 20 \text{ cm.}$$

- 12 Find the perimeter of a sector with radius 4 cm and angle $\frac{\pi}{2}$ radians.

$$\text{Perimeter} = 2r + r\theta = 2(4) + 4\left(\frac{\pi}{2}\right) = 8 + 2\pi \approx 8 + 6.28 \approx 14.28 \text{ cm.}$$

The area of a segment

- 13 A circle has radius 5 cm. A chord subtends an angle of $\frac{\pi}{2}$ radians at the center. Find the area of the minor segment, using segment area = sector area minus triangle area.

$$\text{Sector area} = \frac{1}{2}(5)^2\left(\frac{\pi}{2}\right) = \frac{25\pi}{4}. \text{ Triangle area} = \frac{1}{2}(5)(5)\sin\left(\frac{\pi}{2}\right) = \frac{25}{2}(1) = 12.5. \text{ Segment area} = \frac{25\pi}{4} - 12.5 \approx 19.63 - 12.5 \approx 7.13 \text{ cm}^2.$$

- 14 A circle has radius 6 cm. A chord subtends an angle of $\frac{\pi}{3}$ radians at the center. Find the area of the minor segment, using segment area = sector area minus triangle area.

$$\text{Sector area} = \frac{1}{2}(6)^2\left(\frac{\pi}{3}\right) = 6\pi. \text{ Triangle area} = \frac{1}{2}(6)(6)\sin\left(\frac{\pi}{3}\right) = 18\left(\frac{\sqrt{3}}{2}\right) = 9\sqrt{3} \approx 15.59. \text{ Segment area} = 6\pi - 9\sqrt{3} \approx 18.85 - 15.59 \approx 3.26 \text{ cm}^2.$$

Small-angle approximations

- 15 Use the small-angle approximations $\sin \theta \approx \theta$ and $\cos \theta \approx 1 - \frac{\theta^2}{2}$ to approximate $\sin(0.1)$ and $\cos(0.1)$, to four decimal places.

$$\sin(0.1) \approx 0.1. \cos(0.1) \approx 1 - \frac{0.01}{2} = 1 - 0.005 = 0.9950.$$

- 16 Use the small-angle approximation $\tan \theta \approx \theta$ to approximate $\tan(0.2)$, and compare it to the exact value $\tan(0.2) \approx 0.2027$.

$\tan(0.2) \approx 0.2$ using the small-angle approximation. The exact value is $\tan(0.2) \approx 0.2027$, so the approximation differs by only about 0.0027.

Angular speed and radians per second

- 17 A wheel rotates at 5 radians per second. Find the angle turned through, in radians, after 8 seconds, and convert this to the number of complete revolutions.

$$\text{Angle} = 5 \times 8 = 40 \text{ radians. Number of revolutions} = \frac{40}{2\pi} \approx 6.37 \text{ revolutions.}$$

- 18 A wheel rotates at 3 radians per second. Find the angle turned through, in radians, after 15 seconds, and convert this to the number of complete revolutions.

$$\text{Angle} = 3 \times 15 = 45 \text{ radians. Number of revolutions} = \frac{45}{2\pi} \approx 7.16 \text{ revolutions.}$$

A well-designed word problem: a swinging pendulum's arc

- 19** This question has two parts. A pendulum of length 1.2 meters swings through an angle of 0.3 radians on each side of vertical (a total swing angle of 0.6 radians). (a) Find the length of the arc traced by the pendulum's tip during one full swing (from one side to the other). (b) Find the area swept out by the pendulum during this swing.

(a) Arc length $= r\theta = 1.2 \times 0.6 = 0.72$ meters. (b) Area of the sector swept
 $= \frac{1}{2}r^2\theta = \frac{1}{2}(1.2)^2(0.6) = \frac{1}{2}(1.44)(0.6) = 0.432$ square meters.